Practice workbook · Use aseptic technique to test antibacterial agents on agar

Petri-dish investigations of antibiotics, cleaners or disinfectants: aseptic transfer, contamination control, culture safety and judging effectiveness by clear zones or colonies.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-ASEPTIC-AGAR-PLATE-TEST

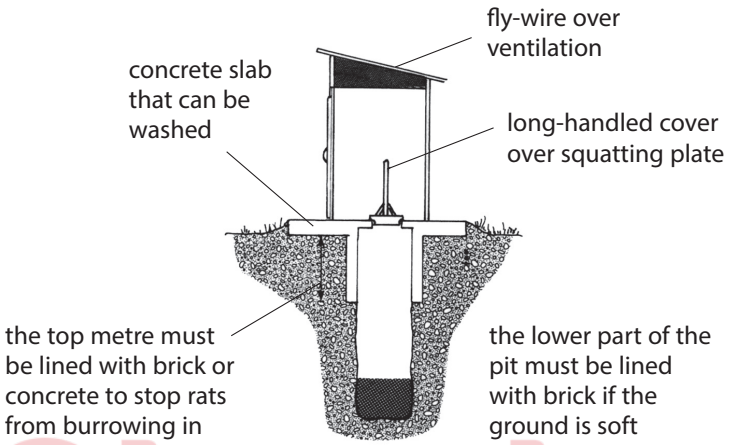
1. Question 5 · part b · 7 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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1. Question 5 · part b · 7 marks

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Worked-example retry · Attempt part b only.

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(iii) Explain why, when a pit latrine is not in use, the latrine should have a cover and why the cover should have a long handle.

(5)

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1. Question 5 · part b · 7 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

(b) The concrete slab shown in the diagram is washed regularly with an antibacterial solution.

A scientist wants to investigate two antibacterial solutions to see which solution is more effective.

Design an investigation the scientist could use to find out which of the two antibacterial solutions is more effective.

Include experimental details in your answer and write in full sentences.

(7)

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(Total for Question 5 = 15 marks)

1. Question 5 · part b · 7 marks · Working and answer

Working and answer

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2. Question 4 · part b.i · 2 marks

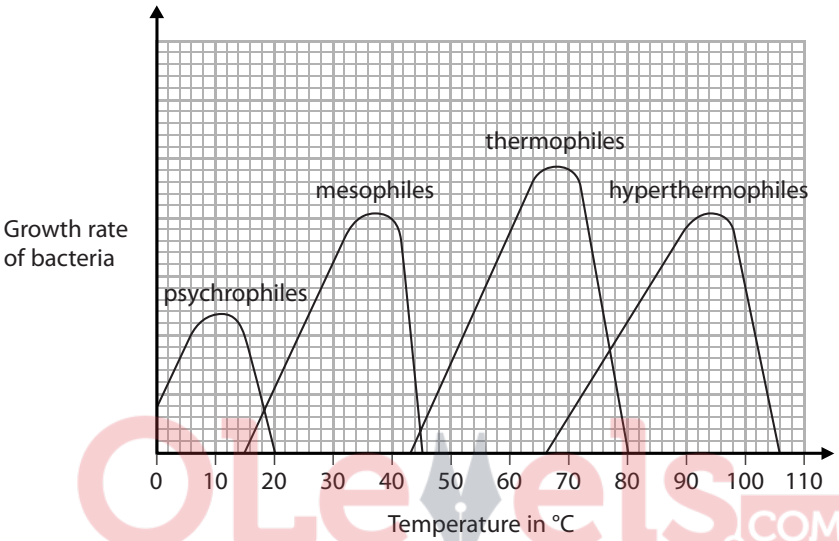
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Further practice · Attempt part b.i only.

4 (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.



(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

(1)

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
- ☒ **D** hyperthermophiles

(ii) Explain how cooking food at 65 °C will affect thermophiles.

(2)

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2. Question 4 · part b.i · 2 marks

June 2019 Paper 1

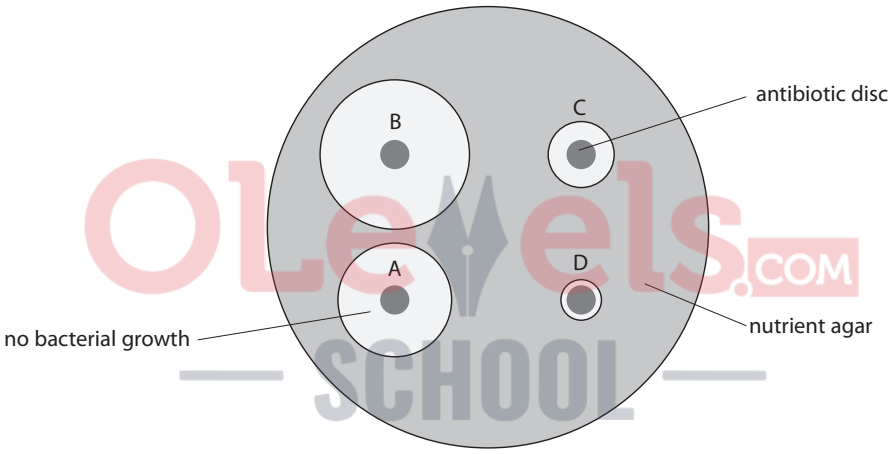
Further practice · Attempt part b.i only.

(b) A student investigates the effect of different antibiotics on the growth of bacteria.

She uses this method.

- streak bacteria onto nutrient agar in a Petri dish
- place discs of different antibiotics, A, B, C and D, onto the nutrient agar
- incubate the Petri dish in a warm oven for one week

The diagram shows the results after one week.



(i) Explain how the student could determine the effectiveness of each antibiotic.

(2)

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2. Question 4 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

(ii) Explain safety precautions the student should take in this investigation.

(6)

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(Total for Question 4 = 11 marks)

2. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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3. Question 4 · part b.ii · 6 marks

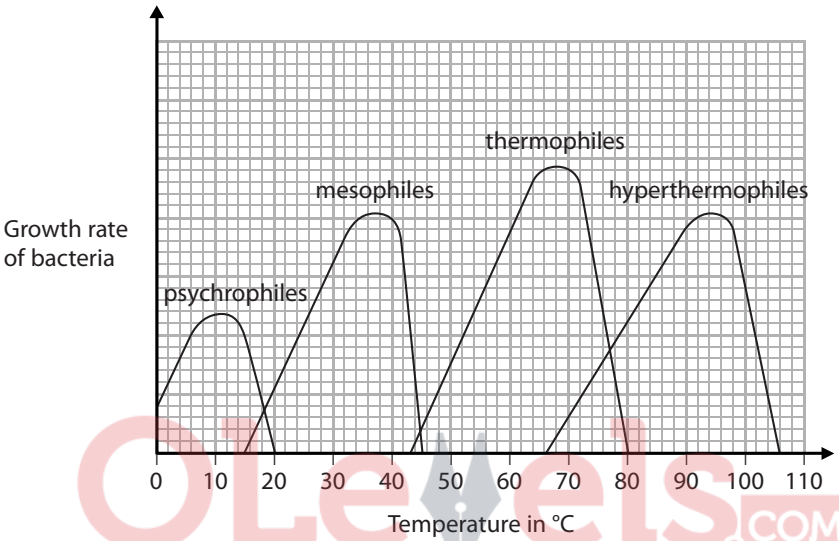
June 2019 Paper 1

Further practice · Attempt part b.ii only.

4 (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.



(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

(1)

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
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(2)

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3. Question 4 · part b.ii · 6 marks

June 2019 Paper 1

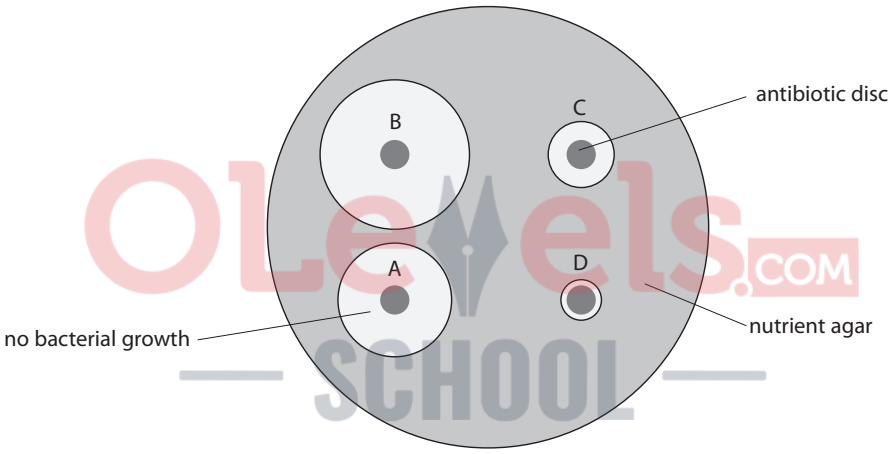
Further practice · Attempt part b.ii only.

(b) A student investigates the effect of different antibiotics on the growth of bacteria.

She uses this method.

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(2)

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3. Question 4 · part b.ii · 6 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(ii) Explain safety precautions the student should take in this investigation.

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(Total for Question 4 = 11 marks)

3. Question 4 · part b.ii · 6 marks · Working and answer

Working and answer

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4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	**MRSA**	***S. aureus***	**MRSA**
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

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4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

(1)

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

(6)

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(Total for Question 6 = 15 marks)

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4. Question 6 · part b.ii · 6 marks · Working and answer

Working and answer

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(ii) State the dependent variable in the student's investigation.

(1)

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

(d) Suggest what further tests the student should do to show that garlic can be used as an effective general antibacterial agent.

(2)

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(Total for Question 7 = 14 marks)

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Working and answer

6. Question 6 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

6 A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

(1)

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

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6. Question 6 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

(2)

(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

(1)

(Total for Question 6 = 6 marks)

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Working and answer

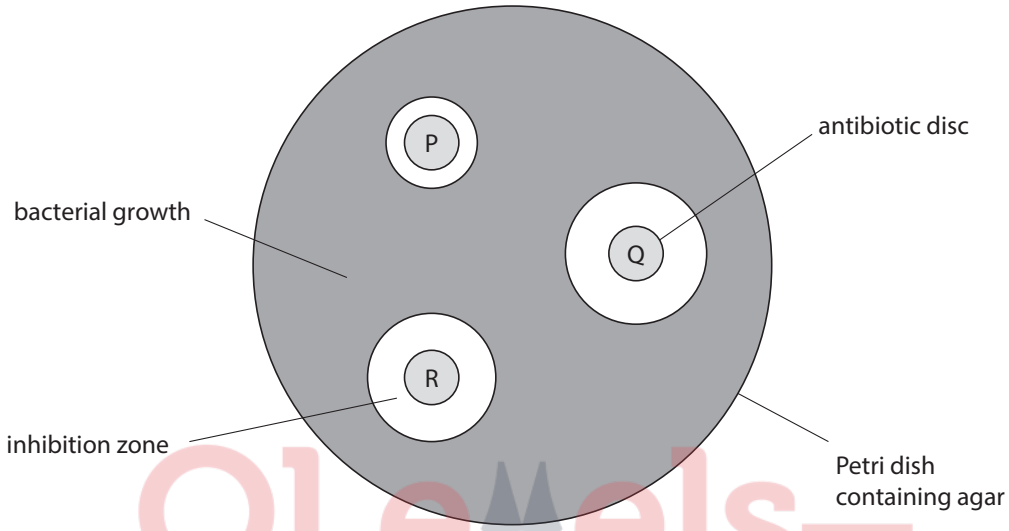
7. Question 5 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

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7. Question 5 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

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7. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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8. Question 6 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

6 A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

(1)

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

(2)

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8. Question 6 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

(2)

(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

(1)

(Total for Question 6 = 6 marks)

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Working and answer